

A Reference Model for AI Enabled Cyber Threat Intelligence Reporting

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ABSTRACT

Artificial intelligence has advanced cyber threat intelligence capabilities in data collection, structuring, and analysis, with mature tooling and strong benchmark results across these tasks. A systematic literature review of research mapped against the SEI Cyber Intelligence Framework reveals that reporting, strategic analysis, and human machine teaming receive significantly less research coverage than collection and analysis. Faster and more accurate collection increases the volume of intelligence without proportionally improving the quality, relevance, or usefulness of what reaches decision makers. This paper proposes a reporting centered reference architecture that stores organizational context, including sector, regulatory environment, and prioritized intelligence requirements, in a retrieval accessible knowledge base. The architecture involves the analyst as a co-equal partner in drafting and reviewing intelligence products, captures structured feedback from each edit session to improve subsequent outputs, and generates three classes of reports from the same evidence base: operational briefs for analysts and responders, strategic briefs for executive leadership, and anonymized products for external sharing communities. The paper also proposes evaluation metrics for gauging the effectiveness of AI assisted intelligence products, contributing a reporting focused evaluation dimension where previous assessments have concentrated on quantified extraction accuracy alone.

Index Terms - Cyber threat intelligence, LLM cybersecurity, threat intelligence reporting, generative AI, retrieval augmented generation, threat intelligence automation, reference architecture, security analyst workflow

I. INTRODUCTION

Cyber threat intelligence exists to inform defenders before incidents occur and to shape how organizations invest in their security posture. A functioning CTI program collects threat data, analyzes it against the organization's specific risk

profile, converts findings into structured reports for different audiences, and feeds consumer response back into future collection priorities. Each of these stages matters. An organization that collects well but reports poorly accumulates data without accumulating understanding.

The past several years have produced a growing body of AI research and tooling aimed at CTI workflows. Multiple surveys document this expansion across extraction, classification, and analysis tasks [5], [13], [15], [33]. Systems now perform well on indicator extraction, ATT&CK technique mapping, and structured data formatting, with established benchmarks and published evaluation results across these capabilities. The research attention, however, has not been distributed proportionally. Collection and analysis have received dense coverage, while reporting, structured feedback, and strategic analysis remain underrepresented.

The consequence of this imbalance compounds as collection tools improve. When AI driven pipelines surface more candidate intelligence faster, but reporting and strategic analysis remain manual and informal, the analyst absorbs the growing volume without proportionally stronger support for the work that matters most: prioritization, synthesis, and communication to varied stakeholders. Most current research scores well on benchmarks that measure extraction precision, recall, and F1.

A CTI system that extracts 10,000 IOCs with 95% precision but generates no usable intelligence product would score well on current benchmarks. Whether it made any organization safer is a question that remains unaddressed.

Whether the resulting output was useful to the organization it was meant to serve, whether it reached the right audience, or whether it generated feedback that improved subsequent collection goes unmeasured. The field evaluates systems against the tasks they were designed for rather than against the outcomes they are supposed to produce.

Research Contribution

This paper proposes a reporting centered reference architecture for AI enabled cyber threat intelligence and contributes the following:

- a) Architecture that treats intelligence product generation as a primary design target, consuming outputs from existing extraction and enrichment tools to produce structured reports.
- b) Organizational context layer that stores sector, regulations, assets, and prioritized intelligence requirements, enabling triage based on relevance.
- c) Three classes of intelligence products generated from the same evidence base: operational briefs for analysts and incident responders, strategic briefs for executive leadership, and anonymized products for external sharing with ISACs and peer organizations.
- d) Designed human machine teaming workflow with explicit task allocation, LLM dependency disclosure, and feedback capture at the analyst, consumer, and system levels that flows back into collection priorities.
- e) Evaluation framework comparing AI only, human only, and human with AI conditions using compliance, product usefulness, and edit distance as metrics.

The question this paper investigates is whether a reporting centered, organization-aware architecture can address the reporting and feedback gap documented in the literature and in practitioner evidence. Can AI assisted, human reviewed intelligence products make senior decision makers more informed about organizational risk? Can anonymization and standardized formatting reduce the friction that currently discourages information sharing?

A. Key Definitions

Cyber Threat Intelligence (CTI) is threat information that has been aggregated, analyzed, and enriched to provide context for organizational decision making.

Cyber Intelligence Framework (CIF) is the six phase lifecycle model defined by the SEI, consisting of environmental context, data gathering, threat analysis, strategic analysis, reporting and feedback, and human machine teaming [1].

Intelligence Products are structured reports with defined consumers, stated confidence levels, source attribution, and actionability requirements.

II. RELATED WORK

A. Prior Work in Cyber Threat Intelligence

The SEI Cyber Intelligence Tradecraft Report, published in 2019 based on structured interviews with 32 U.S. organizations, formalizes the Cyber Intelligence Framework (CIF) as a lifecycle of six related functions: environmental context, data gathering, threat analysis, strategic analysis, reporting and feedback, and human machine teaming [1]. Each function was assessed against maturity indicators derived from Intelligence Community analytic standards (ICD 203) and structured analytic techniques designed to counteract cognitive bias. Among the most extensively documented weaknesses is reporting and feedback.

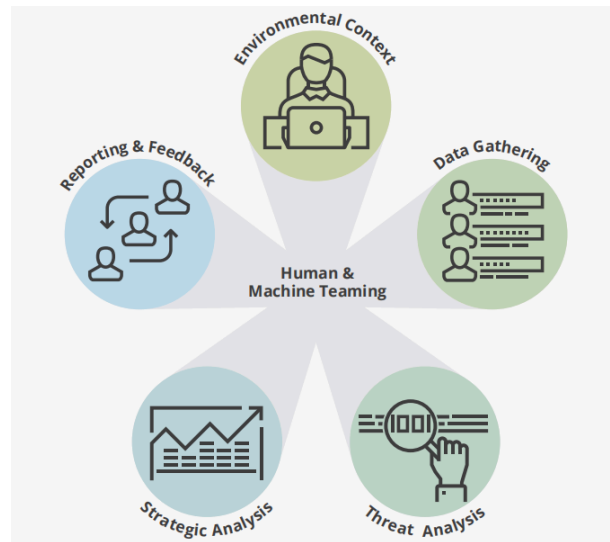


Fig. 1. The SEI Cyber Intelligence Framework [1]

Raw Data to Usable Intelligence

The report observed that many CTI teams distribute intelligence products through informal channels (email, ad hoc briefings, verbal updates) rather than formalized product lines with embedded feedback collection. High performing organizations, by contrast, maintain varied product lines spanning vulnerability reports, threat actor profiles, tactical indicator summaries, executive briefings, and annual threat assessments, each tied to defined audiences, service level agreements, and structured consumer feedback mechanisms [1]. Strategic analysis, defined in the CIF as the holistic assessment of threats, risks, and opportunities for executive decision making, depends on reasoning discipline, uncertainty communication, and traceability of analytic judgments rather than extraction accuracy alone [1].

Cyber Threat Intelligence Disconnect

The cognitive foundations of analytical work in intelligence predate CTI as a discipline. Heuer's *Psychology of Intelligence Analysis* documents how analysts construct mental models that simplify complex realities, how those models introduce systematic biases under conditions of ambiguity, and how techniques such as Analysis of Competing Hypotheses can partially counteract those effects [37]. A central finding is that analysts who received more information became more confident in their existing judgments without becoming measurably more accurate [37].

CTI inherits its conceptual vocabulary from military intelligence, but the translation to commercial organizations is difficult. Kotsias et al. observed that the literature focuses heavily on technical aspects of CTI (event aggregation, correlation, threat hunting, forensic analysis) while remaining largely silent on how commercial organizations can adopt the readiness posture that CTI implicitly demands [6]. Their response was to reconceptualize **CTI as a service**, drawing on nine quality criteria from the U.S. Army's FM 2-0 intelligence manual (accurate, relevant, complete, precise, timely, usable, reliable, predictive, tailored) and applying them to stakeholder driven delivery within enterprise environments [6].

Information sharing introduces additional friction

NIST SP 800-150 establishes a federal standard for the coordinated exchange of indicators, threat actor information, and vulnerability data [3]. Abraham et al. document persistent barriers including trust deficits, misaligned incentives, attribution concerns, and the absence of utilization metrics [4]. The SEI tradecraft report corroborates this at the practitioner level, noting that organizations voiced frustration with ISACs where shared intelligence was described as occasionally negligible or already known before release [1].

MISP represents the most widely deployed implementation of structured CTI sharing [11]. Wagner et al. describe its architecture of machine readable events, correlation engines, graduated distribution scopes, and a proposal mechanism enabling cross organizational correction [10].

The MISP best practices documentation extends this into the analytical domain, urging distinguishing three improvement scenarios: notifying the original analyst of errors, performing complementary analysis for a different consumer group, and producing outputs for automated systems. M. Ammi et al. applied MISP in a threat hunting context, demonstrating a layered architecture connecting structured sharing to active defensive operations [12]. Sánchez del Monte and Hernández-Álvarez analyze across CTI frameworks for strategic, operational, and tactical intelligence tiers [34].

Insufficient CTI Benchmarking

Bjurling and Raza examined whether AI generated CTI outputs meet the analytic standards that tradecraft frameworks demand [7]. Their assessment found that current evaluation benchmarks for AI in CTI measure extraction performance (precision, recall, F1) rather than adherence to intelligence community requirements such as source reliability, confidence calibration, and estimative language [7]. This disconnect between how AI for CTI systems are evaluated and what intelligence consumers need from finished products is a recurring observation across the literature in this paper.

B. AI in Cyber Threat Intelligence

AI capabilities applied to CTI tasks have developed unevenly across the lifecycle. The following subsections organize existing work by capability area.

Extraction and Structuring

Converting unstructured threat reports into structured, machine readable representations is the most mature capability area in AI for CTI research. Multiple working systems exist with published evaluations. STIXnet provides a modular pipeline for extracting STIX-formatted objects from natural language text, integrating extracted entities against an existing knowledge base and incorporating analyst validation for entities not previously cataloged [8]. TSTEM combines focused web crawling with relevance filtering and IOC extraction, designed for continuous deployment against live threat feeds [9]. The ATT&CK framework has become a focal point for structured extraction: S. Zhang et al. modeled an automatic mapping system that scores extracted technique associations for quality and formats outputs for STIX/TAXII distribution [29], while TechniqueRAG addresses the hierarchical complexity of ATT&CK technique annotation, where parent techniques and sub techniques require different levels of contextual evidence to annotate correctly [47]. Two systems push extraction toward richer semantic output. Vulcan, developed by Jo et al., moves beyond traditional IOC extraction to capture descriptive entities and semantic relationships in graph backed representations [30]. CTI Thinker combines knowledge graph construction with GraphRAG reasoning for extraction, alignment, and inference within a single framework [43].

Retrieval, Reasoning, and RAG

A growing body of work applies retrieval augmented generation to CTI tasks, grounding LLM outputs in retrieved evidence rather than relying on parametric knowledge alone. Paul et al. constructed a RAG based system fed by real time CVE, CWE, EPSS, and KEV data sources, enabling prioritization grounded in current vulnerability intelligence [23]. Tellache et al. proposed a dual retrieval architecture combining structured IOC/API lookups with semantic search

over historical CTI documents, using an intermediate alert expansion step that enriches raw SIEM alerts with contextual threat data before generating response plans [24]. Their evaluation employed answer relevance, context relevance, and groundedness as criteria, cross validated against human expert assessments [24]. LocalIntel retrieves both global CTI knowledge from public feeds and local organizational context from internal repositories, producing organization specific implications and recommended counteractions rather than generic summaries [31]. Clouse and Valadez built a human centered RAG system for CTI labeling and forecasting that preserves analyst feedback across retrieval iterations [46]. Hamzić et al. evaluated graph augmented and agentic retrieval approaches against standard RAG for CTI and found that graph grounding improved performance on structured factual questions but introduced schema brittleness and structural hallucination when queries did not align with the graph schema [48].

Lifecycle and Workflow Integration

Several systems attempt to address multiple CTI lifecycle stages within a single architecture. Spyros et al. presented a framework incorporating collection, enrichment, and analysis with MISP integration [2]. Shah et al. built a CTI analysis system on Azure that separates strategic/executive output sections from technical/operational ones, with human validation required before any product is released [17]. Alturkistani et al. modeled CTI workflows covering topic categorization, extraction, TTP classification, STIX mapping, and mitigation output generation, with explicit separation of strategic, operational, tactical, and technical CTI tiers [32]. Tseng et al. proposed an LLM based system targeting end to end CTI processing from report ingestion to structured SOC consumable outputs [21]. Among these, Shah et al. [17] and Alturkistani et al. [32] are notable because they produce differentiated output formats for different consumer audiences, a design pattern that recurs in this paper's proposed architecture.

Summarization and Intelligence Product Generation

Balasubramanian et al. conducted the largest systematic review to date, analyzing 330 papers on AI applications in CTI and mapping research activity across CTI functions [5]. Ding and Lu documented a knowledge guided summarization approach where domain specific CTI terminology, injected as input, improved factual consistency and reduced hallucination rates compared to generic summarization baselines [19]. The CTISum benchmark, developed by Peng et al., targets downstream CTI summarization tasks including attack process summarization and evaluates current models against them [35]. A systematic survey of text summarization [41] documented evaluation criteria applicable to intelligence product generation: faithfulness to source material, factual consistency, controllability of output format, audience

adaptation, and position bias where LLMs systematically underweight information appearing in the middle of long input contexts. Two recent multi agent approaches address the complexity of synthesizing intelligence from multiple sources. A multi agent pipeline for threat intelligence summarization [51] handles IoC extraction, dynamic behavior analysis, MITRE technique mapping, and contradictory evidence across sources. A knowledge graph based multi agent approach to CTI [52] accommodates ambiguity, supports abstention when evidence is insufficient, and allows multiple acceptable interpretations of the same threat data.

C. Broader AI in Cybersecurity

Research on AI in cybersecurity beyond CTI provides context for how LLMs interact with security workflows more broadly. A survey of LLMs for cyber threat detection [13] found that the dominant operational use cases are alert triage, log interpretation, and incident summarization, with BERT-family models prevalent for named entity recognition and relation extraction while GPT-family models dominate generation tasks. Habibzadeh et al. mapped LLM applications across SOC workflows and documented a pattern where research attention is strongest for detection and monitoring and declines as tasks move toward response, communication, and decision support, with trust, explainability, and human oversight identified as recurring requirements for operational deployment [16]. Ferrag et al. reviewed generative AI across cybersecurity and documented dataset lifecycle challenges, hybrid annotation methods, and LLM vulnerability categories affecting deployment reliability [15]. Kaur et al. conducted a systematic review of 236 primary studies and classified research against the NIST Cybersecurity Framework, finding AI maturity clustering around detection, monitoring, and extraction [33]. Zhang et al. mapped LLMs to cybersecurity applications broadly [14], and Saddi et al. examined generative AI's role in threat intelligence enhancement [36].

LLMs for SOC Workflows in Production

Production deployments provide a different view of these capabilities under operational constraints. Microsoft's Security Copilot for Threat Intelligence integrates promptbooks, actor profile generation, vulnerability assessment, IOC extraction, hunting query generation, and summarization into an interface with feedback mechanisms through which analysts can rate and refine generated outputs [49]. Singh et al. studied how SOC analysts use LLMs in operational settings and observed that analysts employ them primarily for context building, sensemaking, and writing support through short, targeted interactions [50]. Analysts consistently retained decision authority over final judgments, treating LLM outputs as starting material rather than finished products [50].

III. METHODOLOGY

This paper employs a systematic literature review (SLR) as its primary research method, supplemented by a lifecycle based gap analysis and the development of a proposed reference architecture. The methodology is structured to answer two connected questions: where does current AI capability map onto the CIF, and where do gaps in that mapping create operational weaknesses that a reporting centered architecture could address.

Search Strategy

The literature search was organized into three thematic buckets, each with distinct keyword combinations. The first bucket targeted prior work in CTI tradecraft and practice, using terms such as *cyber threat intelligence lifecycle*, *CTI reporting*, *intelligence sharing*, *analytic tradecraft*, *structured analytic techniques*. The second targeted AI applied to CTI tasks specifically, using terms such as *LLM cyber threat intelligence*, *NLP threat report*, *automated CTI*, *RAG threat intelligence*, *ATT&CK extraction*. The third targeted broader AI applications in cybersecurity, using terms such as *LLM SOC*, *generative AI cybersecurity*, *AI security operations*. Searches were organized between 2016 and 2026 conducted across Google Scholar, IEEE Xplore, ACM Digital Library, and arXiv.

Inclusion and Exclusion Criteria

Inclusion criteria required that papers directly address CTI processes, AI or LLM applications in security contexts, or adversarial AI dynamics with CTI implications. Papers were excluded if they focused exclusively on general NLP benchmarking without security application, malware binary analysis without connection to CTI lifecycle functions, or predated 2019, the publication year of the SEI Cyber Intelligence Tradecraft Report [1] that serves as this paper's analytical baseline. Exceptions were made for foundational works published earlier that remain directly relevant to CTI tradecraft and cognitive analysis, such as Heuer [37].

Corpus Selection and Classification

The initial search yielded approximately 120 candidate papers. After title and abstract screening, 72 were retained for full text review. Of these, 54 were included in the final corpus based on direct relevance to at least one CIF phase or to the design considerations of a reporting centered CTI architecture. This winnowing process follows the general approach documented by Balasubramanian et al. in their 330 paper systematic review of AI in CTI [5], though at a smaller scale appropriate to the scope of this work.

Each paper in the final corpus was classified against the six CIF phases defined in [1]. Where a paper addressed multiple

phases, it was assigned to all applicable phases. Coverage strength was rated using two levels: strong (the paper presents implemented tooling or empirical evaluation for that phase), or moderate (the paper proposes a method or discusses the phase substantively). This classification produced the lifecycle coverage matrix presented in Section IV.

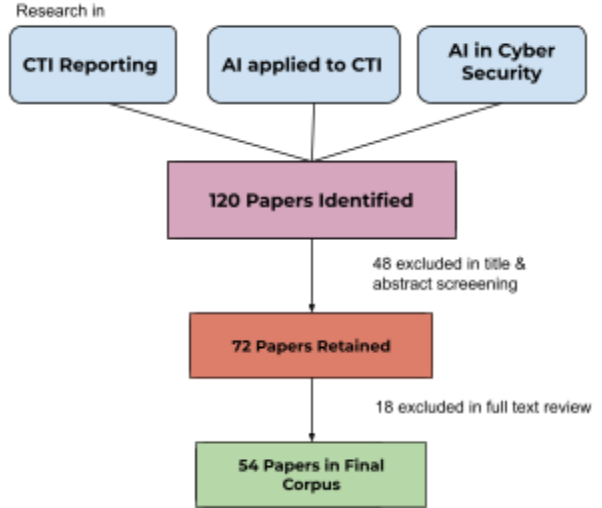


Fig. 2. Literature search and filtering process.

IV. ANALYSIS

The literature reviewed in Section II documents a field with strong and growing capability in certain CTI functions and persistent weakness in others. This section examines where that capability concentrates, what consequences the imbalance produces, and why current evaluation approaches fail to surface the problem.

A. Uneven Coverage Across the CIF

When the 54 papers in the corpus are classified against the six CIF phases, the distribution is not uniform. Data gathering and threat analysis receive the densest coverage. Multiple working systems exist for IOC extraction [8], [9], STIX structuring [8], [29], ATT&CK technique mapping [9], [29], [47], knowledge graph construction [43], and retrieval augmented reasoning over CTI corpora [23], [24], [31], [46], [48]. These systems have published evaluation results, established benchmarks, and in several cases production deployments. Environmental context receives moderate coverage, primarily through organizational context retrieval [31] and intelligence requirement alignment [1], though fewer systems operationalize it as a distinct design layer.

Strategic analysis, reporting and feedback, and human machine teaming receive substantially less coverage. Of the papers that address reporting, most treat it as a byproduct of earlier processing stages rather than a primary design target. Shah et al. [17] and Alturkistani et al. [32] produce differentiated output formats for different consumer audiences, but the majority of systems reviewed terminate at extraction, enrichment, or retrieval without generating intelligence products suited to specific audiences. By the CIF's own standard, a system that extracts perfectly but never produces a consumable product has not completed the lifecycle [1]. Balasubramanian et al.'s 330 paper meta-analysis confirms this pattern at a larger scale, documenting that research activity in AI for CTI clusters around collection and analysis while reporting and strategic functions receive disproportionately less attention [5].

B. Consequences of Coverage Imbalance

The uneven distribution documented in Section IV.A has operational consequences that compound as collection tools improve. When extraction and enrichment pipelines become faster and more voluminous but reporting and strategic analysis remain manual and informal, the system generates more candidate intelligence without proportionally improving its quality, relevance, or usefulness to decision makers.

The SEI tradecraft report frames this directly: cyber intelligence should be guided by intelligence requirements and organizational priorities, not by collection volume alone [1]. The report identifies the absence of an organization-wide intelligence requirement process as a common challenge and notes that stale or poorly defined requirements produce irrelevant data collection and diminishing analytical returns.

Without requirements that reflect current organizational priorities, collection becomes self-justifying. More data is gathered because the pipeline can gather it, regardless of its effectiveness and usability.

Distribution of research coverage across CTI lifecycle phases

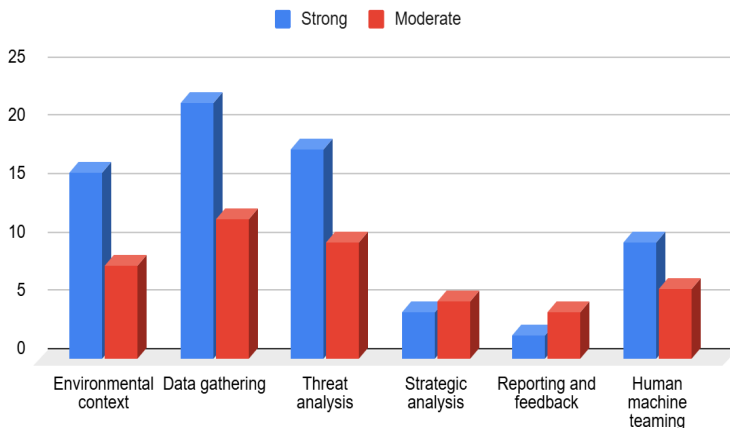


Fig. 3. Cyber Threat Intelligence Lifecycle Phases

This dynamic is visible at the analyst level. As collection tools surface increasing volumes of candidate artifacts, the analyst faces a growing triage burden without proportionally stronger support for prioritization, synthesis, or evaluation. Heuer's cognitive load research [37] and Kotsias et al.'s organizational adoption findings [6] both point to the same conclusion: volume without structure creates overhead, not insight. Organizations that invested in collection infrastructure without corresponding investment in how intelligence reaches stakeholders and whether it shapes decisions saw limited returns on their programs [6].

C. Evidence of Reporting Weakness

The pattern visible in the literature review is corroborated by benchmark results. Several recent benchmarks have evaluated LLM performance across CTI task types, and the results consistently show degradation as tasks move from structured extraction toward reasoning and product generation.

CTIBench evaluates models on threat intelligence knowledge recall, severity prediction, IOC extraction, and mitigation recommendation [22]. Models perform well on extraction and knowledge recall but struggle with mitigation, where the task requires synthesizing threat context into actionable guidance rather than identifying known patterns. AthenaBench introduces dynamic evaluation across CTI tasks and documents similar degradation: models are weakest on mitigation recommendation and actor attribution, tasks that require contextual reasoning rather than pattern matching [27]. The Reasoning over Mitigative Strategies task in AthenaBench is particularly relevant because it evaluates exactly the kind of analytical synthesis that strategic intelligence products demand.

CTIArena evaluates models across heterogeneous CTI tasks spanning structured, unstructured, and hybrid formats [42]. The results show that synthesis across multiple sources, where the model must reconcile information from several reports to produce a coherent assessment, is significantly harder than extraction from a single source. The CTISum benchmark specifically targets summarization of CTI content and finds that current systems produce imperfect results even on relatively constrained summarization of attack processes [35].

CyberThreat-Eval takes an evaluation approach centered on analyst tasks that most closely mirrors real CTI workflows [44]. Rather than testing extraction in isolation, it evaluates whether models can perform realistic analyst work from end to end, including TTP reasoning and actor profiling. Reporting and assessment tasks proved consistently more difficult than extraction tasks [44].

Mezzi et al. evaluated LLM reliability on realistic, complete CTI reports rather than short curated excerpts and found that performance degrades substantially as input length and complexity increase [28]. Models exhibited inconsistency across repeated evaluations of the same material and produced overconfident assessments without appropriate hedging or uncertainty language [28]. For any architecture that relies on LLMs to process real CTI feeds, which are rarely short or neatly structured, this finding constrains how much trust can be placed in unreviewed generated output.

D. Strategic Analysis and Reporting as the Key Gap

The coverage matrix, the benchmark evidence, and the practitioner data converge on the same conclusion. Strategic analysis and reporting are the phases most necessary for converting intelligence work into organizational value. They are also the phases where current AI capability is weakest.

The SEI tradecraft report defines strategic analysis as producing assessments that help leadership make decisions about organizational risk posture, resource allocation, and operational priorities [1]. This requires estimative reasoning, uncertainty communication, and the ability to frame threat activity in terms that non-technical decision makers can act on. None of the extraction, retrieval, or enrichment tools reviewed in Section II produce this kind of output. The closest precedents are Shah et al.'s system [17], which separates strategic from operational output sections, and Alturkistani et al.'s workflow [32], which distinguishes strategic, operational, tactical, and technical CTI tiers. Even these systems focus primarily on structuring and categorizing rather than on producing intelligence products that meet the tradecraft standards the SEI report describes.

The Threat Intelligence Benchmark provides evidence from practitioners that confirms what the research literature suggests. Enterprise security teams report being overwhelmed by threat data and struggling to convert collected intelligence into actionable products that reach the right stakeholders at the right time [45]. The bottleneck sits at contextualization, prioritization, and communication.

Singh et al.'s empirical study of LLM use in SOC environments adds a behavioral dimension [50]. Analysts already use LLMs for context building, sensemaking, and writing support. They consistently retain decision authority over final judgments, treating generated output as a starting point for further work rather than a finished deliverable [50]. This observed behavior is itself evidence of where AI fits within the intelligence workflow: as a production tool that accelerates drafting and structuring before human review,

while analytical judgment stays with the analyst. No current system formalizes this collaboration pattern into a designed workflow with structured feedback capture.

E. The Evaluation Disconnect

A subtler problem runs beneath the coverage and capability gaps. The way AI in CTI is evaluated does not measure what matters most to intelligence consumers.

Current evaluation operates at the task level. Papers report precision, recall, and F1 on extraction tasks. They report BLEU, ROUGE, or BERTScore on summarization tasks. These metrics describe whether the system correctly identified an IOC or produced a summary that overlaps with a reference text. They do not describe whether the intelligence product was useful to the organization it was intended to serve, whether it reached the right audience, whether it changed a decision, or whether it generated feedback that improved subsequent collection.

Bjurling and Raza identified this disconnect directly: benchmarks measure extraction performance rather than adherence to intelligence community standards such as source reliability, confidence calibration, and estimative language [7]. The evaluation criteria proposed by Tellache et al. (answer relevance, context relevance, and groundedness) [24] and the faithfulness, factual consistency, and audience adaptation criteria documented in a systematic survey of text summarization [41] represent steps toward evaluation that is relevant to intelligence production, but neither has been widely adopted as a standard.

Meng et al. mapped three categories of LLM failure specific to CTI tasks: spurious correlations where models associate threats based on surface similarity rather than causal evidence, contradictory knowledge where models reproduce conflicting claims from different sources without reconciliation, and constrained generalization where models fail to reason about novel threats that differ from training distributions [20]. These failure modes compound across a pipeline. An extraction error in data gathering becomes a false correlation in threat analysis, which becomes a misleading claim in a strategic brief. Evaluation at the task level catches the extraction error. It does not catch the cascading consequences.

The result is a field where systems are measured against the tasks they were designed for rather than against the outcomes they are supposed to produce.

V. RECOMMENDATIONS & PROPOSED ARCHITECTURE

The analysis in Section IV identifies strategic analysis, reporting and feedback, and human machine teaming as the CIF phases with the weakest AI coverage and the highest operational consequence. The architecture proposed here addresses those phases directly. It does not attempt to replace or replicate the extraction, enrichment, and retrieval systems documented in the literature. Those capabilities are mature and improving. The architecture sits after them, consuming their outputs and producing intelligence products that close the lifecycle loop.

The architecture is governed by three design constraints:

It must address the phases where the lifecycle is least complete. All components must be technically feasible using existing tools and documented methods. The analyst must remain a co-equal partner in intelligence production.

A. Reporting Centered CTI Workflow

The architecture treats intelligence product generation as a primary design target rather than a byproduct of data gathering. In the CIF, reporting and feedback is the mechanism through which intelligence becomes visible to operational teams and leadership, and through which new intelligence requirements and coverage gaps are identified [1]. Shah et al. [17] and Microsoft's Security Copilot [49] both demonstrate that AI assisted intelligence product generation is technically feasible. Neither provides an architecture where reporting is designed around organizational context, tradecraft standards, and structured feedback from the start. The architecture proposed here addresses that absence.

The workflow operates in five stages. Structured CTI data enters the system from MISP, feeds compatible with STIX, internal threat intelligence platforms, or manual analyst input. Enrichment is applied through ATT&CK mapping, source retrieval, and artifact normalization using existing tools documented in [8], [9], [29], [47]. An organizational context layer filters and prioritizes the enriched data against the consuming organization's intelligence requirements, sector, and regulatory environment. The LLM generates draft intelligence products from the filtered and contextualized evidence base. The analyst reviews, edits, and approves the draft before release, and the system captures the review process as structured feedback data. Each stage is modular. Organizations with existing extraction pipelines can connect at stage three without rebuilding what came before.

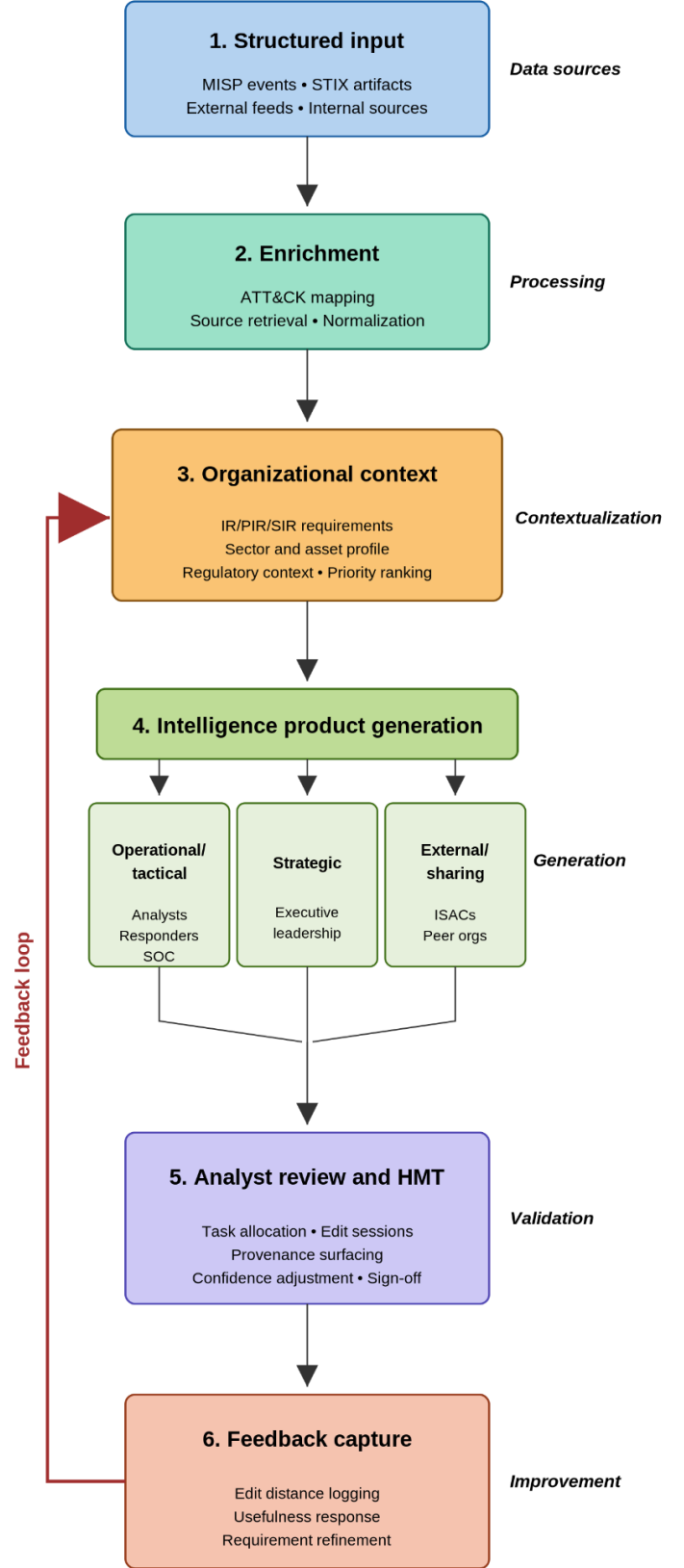


Fig. 4. Reference architecture pipeline

B. Organization Aware Triage and Intelligence Requirements

Generic CTI is not useful until it is contextualized against the consuming organization's infrastructure, operational state, threat exposure, and business priorities. The SEI tradecraft report identifies the absence of a formalized intelligence requirement process as one of the most common weaknesses across the organizations it assessed [1]. Requirements that are stale, vague, or disconnected from business objectives produce collection with limited utility and products with no audience.

1. The architecture addresses this through a structured organizational context layer. When the system is first configured for an organization, an analyst completes a context questionnaire.
2. The questionnaire covers industry sector and sub sector, critical assets and infrastructure dependencies, applicable regulatory frameworks (NIST RMF, PCI DSS, HIPAA, FERPA, and others relevant to the operating environment), known threat actor targeting relevant to the sector, current monitoring coverage and known gaps, and prioritized intelligence requirements at the IR, PIR, and SIR levels.
3. This profile is stored in a retrieval accessible knowledge base and maintained as a living document that analysts update as the organization's priorities, assets, and threat landscape change.
4. When new CTI data enters the pipeline after enrichment, the retrieval layer queries this organizational context alongside the structured threat data and supporting documents.
5. The result is triage informed by context: findings are ranked and filtered based on whether they are relevant to what the organization actually cares about, not just whether they are technically interesting.
6. Before intelligence product generation begins, the analyst is presented with a screen of recommended data sources and contextual matches and selects which to include. This preserves analyst agency over what enters the product while offloading assessment to the system.

LocalIntel provides the closest precedent in the literature for this approach. Mitra et al. demonstrated that combining global CTI knowledge from public feeds with local organizational context from internal repositories produces implications and recommended counteractions specific to the organization rather than generic summaries [31]. The dual retrieval architecture proposed by Tellache et al., which combines structured IOC and API lookups with semantic search over historical CTI documents, provides a complementary technical pattern for the retrieval layer [24].

After intelligence product generation, the system evaluates the draft product against the organizational context profile a second time. This validation step checks whether the generated product addresses the intelligence requirements that initiated the query, whether the prioritization reflects the IR/PIR hierarchy configured in the profile, and whether relevant organizational context (sector exposure, asset criticality, regulatory implications) is represented in the output. If the product fails to meet these thresholds, the system flags the gaps for the analyst before the review session begins. This ensures that organizational relevance is enforced at both the input and output stages of the pipeline rather than only during initial triage.

C. Three Classes of Intelligence Products

The architecture generates three classes of intelligence products from the same structured evidence base. Each class serves a different consumer audience, carries different confidence and formatting requirements, and fills a different role in the organizational intelligence cycle. The distinction between classes reflects the SEI tradecraft report's finding that high performing organizations maintain varied product lines rather than compressing all output into a single undifferentiated summary [1]. The U.S. government's Cyber Threat Intelligence Integration Center publishes multiple product types organized by audience and decision horizon, providing a practical example of this product line approach in operation [53].

Throughout this section, "intelligence products" refers to the structured reports generated by the architecture. The term is used to emphasize that these reports carry tradecraft requirements (source attribution, confidence calibration, audience awareness) beyond standard document formatting, but the deliverable itself is a report.

Operational and Tactical Products

These products are intended for analysts, threat hunters, incident responders, and SOC personnel. They are the most technically detailed of the three classes and the closest to current CTI output formats. The output format is consistent with what Bjurling and Raza describe as the minimum requirements for tradecraft compliant products: source attribution, confidence, and evidence traceability [7].

Operational / Tactical Brief	
For SOC, IR, threat hunters, analysts	
Threat	JADE SERPENT exploiting ActiveMQ...
Evidence	C2 IPs, beacon hash, CISA + FS-ISAC...
ATT&CK	T1190, T1059.001, T1003.001...
Assets	PROD-MQ-01/03, payment message bus
Relevance	Two unpatched PCI-scope ActiveMQ nodes
History	Prior ActiveMQ patch sprint; no exploit seen
Actions	Patch in 24h, restrict 8161, sweep C2...
Gaps	Exfil channel low confidence; peer access...
Review: analyst sign-off required	

Fig. 5. Operational Brief

Each operational product contains: a threat summary describing the incident, campaign, or threat cluster in concise technical language; a list of relevant indicators and artifacts with source attribution; mapped ATT&CK techniques with confidence ratings for each mapping; an organizational relevance statement explaining why this finding matters to this specific organization based on the context profile; a list of affected or potentially affected assets drawn from the organization's asset inventory; related historical activity surfaced from the organization's internal repository showing whether similar threats have been observed before; recommended immediate actions or investigative next steps; and open questions that the analyst may want to pursue.

Strategic Products

These products are intended for executive leadership, board level briefings, and planning functions. They are narrower, more selective, and more explicit about confidence, uncertainty, and organizational impact than operational products.

Strategic Brief	
For executives, CISO, board, risk owners	
Bottom line	Two unpatched ActiveMQ systems at risk
Business impact	Payment routing and settlement exposure
Sector relevance	Financial messaging campaign expanding...
Confidence	High active exploit; moderate org scope
Decision need	Emergency patch vs. throughput reduction
Compliance	PCI DSS 6.3.3, NIST SI-2, SOX risk
Intel gaps	Initial access and lateral movement unknown
Posture	Override 30-day cycle; assess compromise...
Review: analyst + leadership validation	

Fig. 6. Strategic Brief

Each strategic product contains: an executive summary of no more than two paragraphs stating what happened, why it matters to this organization, and what decision it requires; a sector and business significance section that translates the threat into language relevant to the organization's industry, market position, or operational dependencies; a confidence statement using estimative language that explicitly states what is known, what is assessed with moderate or low confidence, and what remains unknown; identified intelligence gaps where collection or analysis could not answer a question the requirement asked; compliance implications mapping the finding to relevant regulatory controls; and decision points for leadership framed as questions or options

A strategic product translates threat activity into questions that leadership can act on: whether a campaign materially affects the organization's risk exposure, whether a threat actor should be elevated in monitoring priority, whether an emerging capability changes the investment calculus for a business unit, or whether a pattern requires coordination across departments. This is consistent with the SEI definition of strategic analysis as holistic assessment of threats, risks, and opportunities for executive decision making [1]. The nine quality criteria Kotsias et al. drew from the U.S. Army's FM 2-0 manual (accurate, relevant, complete, precise, timely, usable, reliable, predictive, tailored) serve as a design checklist for this product class [6].

External and Sharing Products. These products are intended for ISACs, peer organizations, and information sharing communities. They present the intelligence in a form suitable for distribution beyond the originating organization. This class requires two capabilities that the other product classes do not.

External / Sharing Product	
For ISACs, peer organizations, partners	
Summary	Anon. ActiveMQ exploitation attempt...
Indicators	C2 IPs, domains, beacon hash defanged
ATT&CK	T1190, T1059.001, T1071.001...
Sector relevance	Financial orgs running ActiveMQ 5.15-5.17
TLP	AMBER+STRICT; FS-ISAC members only
Source handling	Hostnames, ranges, asset IDs removed
Format	MISP event + STIX-ready fields
Feedback	Request sightings, corrections, enrichment
Review: legal, privacy, sharing rules	

Fig. 7. External Brief

First, formatting structured for machine ingestion, compatible with MISP event structures and STIX representations so that receiving organizations can ingest the intelligence into their own platforms [10]. Each external product maps to a MISP event containing typed attributes, tagged with relevant

taxonomies and galaxy clusters, and scoped to the appropriate distribution level (community, connected communities, or public).

Second, anonymization. The system strips details that identify the originating organization: asset names, internal infrastructure references, network ranges, internal ticket numbers, and attribution markers. What remains is the threat intelligence itself, the indicators, techniques, behavioral patterns, and sector relevance, preserved in full while the organizational fingerprint is removed. The SEI tradecraft report documents that organizations voiced frustration with sharing environments where meaningful participation was limited by attribution concerns [1]. Abraham et al. confirm that willingness to share increases when contributors can participate without being identified as the source [4]. The architecture addresses this by applying redaction aware of the Traffic Light Protocol as a feature of the external product generation pipeline, enabling sharing to ISACs and MISP communities with appropriate distribution controls at the push of a button rather than through a manual sanitization process.

D. Compliance Context Mapping

None of the 54 papers in the corpus map CTI findings to regulatory compliance frameworks as part of the intelligence product itself, despite the operational demand for this capability in regulated industries. CTI consumers in regulated industries need to know whether a threat has implications for specific controls they are required to maintain.

The architecture's reporting layer maps identified vulnerabilities and threat activity to relevant compliance frameworks alongside ATT&CK technique mappings. When a finding is identified, the intelligence product includes which control frameworks are implicated (NIST RMF, PCI DSS, HIPAA, FERPA, SOX, or others configured in the organizational context profile), which specific controls may be affected, and what compliance reporting obligations the finding may trigger. This turns a threat alert into a product that security, compliance, and risk teams can all act on without requiring a separate translation step.

As this research observed, an LLM processing CTI data can identify that a weakness exists, how to remediate it, and that the weakness contributes to a possible violation of specific regulatory controls. Static analysis tools already produce findings and suggested fixes. An AI enabled CTI system can add the compliance dimension: this weakness carries implications for RMF posture, PCI audits, and HIPAA obligations. That additional layer of meaning is what turns a technical finding into an intelligence product that reaches stakeholders beyond the security team.

E. Human Machine Teaming (HMT) in the Architecture

The architecture is designed as a collaborative workflow where AI and analyst each contribute distinct capabilities, and where the collaboration itself is instrumented for improvement. This is the operational meaning of human machine teaming in the context of this paper.

The HMT literature reviewed in this paper provides the design vocabulary. The Navy HAT Roadmap frames human autonomy teaming as a practical adoption problem with specific barriers: trust, explainability, task allocation, and uncertainty communication [38]. Tuckman's team development model, applied to human AI teams by researchers studying autonomous systems, treats disagreement and friction between human and AI as a developmental stage (storming) rather than a failure condition [39]. The Human AI Collaboration Survey distinguishes three levels of collaboration (tool, assistant, teammate) and documents that each level requires different design decisions around role allocation, oversight, and shared accountability [40]. Singh et al.'s empirical study of SOC analysts confirms that in practice, analysts already treat LLMs as drafting assistants rather than autonomous agents, retaining decision authority while using AI outputs as starting material [50]. The architecture operationalizes these findings through five mechanisms.

Task allocation. The AI system drafts, enriches, structures, retrieves, and generates candidate intelligence products. The analyst validates claims against source evidence, edits for accuracy and completeness, adjusts confidence language, adds contextual judgment that the model cannot access, and signs off before any product is released. AI does not publish intelligence products without human review. This functions as a design constraint from the workflow's genesis.

Structured Edit Sessions. Analyst edit sessions are structured. After the system generates a draft product, the analyst enters a review mode where edits are tracked at the section level. Which sections were modified, which claims were removed, what evidence was added manually, how much effort was required before the product was acceptable. These signals feed directly into the feedback layer described in Section V.F.

Traceable Reasoning. Confidence levels, source provenance, and evidence chains are presented alongside generated content so that analysts can evaluate not just what the system concluded but why. Heuer's work on cognitive bias is relevant here: analysts are better equipped to counteract model errors when they can see the evidence trail rather than receiving a

finished product with hidden reasoning [37]. The Navy HAT Roadmap puts this simply: AI needs to show its work [38].

LLM Dependency Disclosure. Each product carries metadata, which models were used in its generation, what training data limitations those models carry, and where known biases may affect the output. This is not standard practice in any system reviewed in the literature, but it follows logically from the tradecraft requirement that intelligence products be transparent about their sources and methods.

Defensive Design. Defensive principles (role based access control, zero trust segmentation, no credential bypass between modules) are built into the architecture rather than treated as deployment considerations. An AI enabled CTI system that processes sensitive organizational data and generates intelligence products for executive consumption is an attractive target. The architecture accounts for that.

F. Feedback as Structured System Input

The CIF treats reporting and feedback as a single lifecycle phase because intelligence products are valuable only if they shape future collection, analysis, and prioritization [1]. Most AI in CTI systems treat feedback as incidental rather than as a system function. The architecture proposed here treats it as structured input.

Feedback is captured at three levels. At the analyst level, the system records how draft products are revised during the edit session described in V.E. Edit distance between the AI generated draft and the analyst approved final version provides a quantitative signal for how much human correction was needed. Revision density by section identifies which parts of the product the model gets right and which it consistently gets wrong. Claims that analysts remove, evidence that analysts add manually, and confidence language that analysts adjust all become structured data that the system can use to identify its own weaknesses over time.

At the consumer level, each intelligence product includes a lightweight feedback mechanism. Consumers indicate whether the product was useful, whether it raised follow up questions, whether it changed a decision or shifted a priority, and whether it reached them at the right time. This mechanism follows the SEI tradecraft report's documentation of high performing organizations that append surveys to finished products and maintain dedicated engagement teams to capture consumer response [1]. The MISP best practices documentation describes analogous mechanisms at the sharing level: proposals for cross organizational correction, sightings for credibility tracking, and alternate analyses that offer competing interpretations of the same data [11].

At the system level, feedback data flows back into the organizational context layer and the prioritization logic. If consumers consistently report that a certain category of product is not useful, the system adjusts its relevance filtering. If analysts consistently remove a certain type of claim, the system learns to flag rather than assert. If a product generates follow up questions, those questions become candidates for new intelligence requirements. This closes the CIF lifecycle loop by feeding reporting outcomes back into environmental context and collection priorities.

The feedback architecture also generates evaluation data relevant to the three condition comparison described in Section VI. Edit distance, revision density, consumer usefulness scores, and time to approval are all metrics that can be compared across AI only, human only, and human plus AI conditions.

VI. PROTOTYPE DESIGN & EVALUATION

The reference architecture is intentionally scoped so that a meaningful portion can be prototyped and evaluated as a proof of concept. The aim is to test whether a reporting centered, organization aware workflow produces more useful intelligence products than workflows that stop at extraction and retrieval, and to document where the approach fails.

A. Prototype Scope

The prototype is built around a constrained question: when structured CTI data is combined with organizational context retrieval and an LLM based reporting layer, does the resulting intelligence product outperform what an extraction heavy workflow produces on its own? And does adding analyst review to the AI generated draft produce a better product than either AI or human working independently?

The prototype implements a simplified version of the five stage pipeline described in Section V.A. In the first stage, CTI reports, advisories, or threat events are ingested from selected sources and represented in a structured form compatible with MISP event structures or STIX inspired schemas [10], [8]. Real world CTI reports from public sources serve as input data, including published APT campaign analyses, vendor threat advisories, and government cybersecurity alerts.

The second stage applies enrichment through ATT&CK technique mapping and source retrieval using methods documented in [9], [29], [47]. The third stage applies a simplified organizational context layer: a pre configured profile representing a hypothetical organization with defined

sector, critical assets, regulatory obligations, and intelligence requirements. This profile is stored in a RAG accessible format and queried during the retrieval step. The fourth stage uses an LLM to generate draft intelligence products in two formats: an operational/tactical brief and a strategic brief, both structured according to the templates described in Section V.C. The fifth stage captures analyst review through tracked edits, producing the structured feedback data described in Section V.F.

The prototype does not implement the compliance context mapping layer in full. It does implement all three intelligence product classes: operational/tactical briefs, strategic briefs, and anonymized external sharing products. The anonymization pipeline applies a simplified version of the redaction process described in Section V.C, stripping organizational identifiers from the evidence base before generating the external product. Full compliance context mapping, where findings are connected to specific regulatory controls, is described in the architecture as a design recommendation and scoped for future implementation.

B. Evaluation Strategy

The evaluation compares three conditions applied to the same source data and the same organizational context profile.

Condition 1: AI only

The LLM generates intelligence products from the enriched, contextualized evidence base without any analyst involvement. The system produces both operational and strategic briefs autonomously. These products are evaluated as is, with no human editing.

Condition 2: Human only

An analyst produces intelligence products manually from the same source data, with access to the same organizational context profile and the same enrichment outputs. The analyst receives no AI generated draft. This condition establishes a baseline for what a trained human produces given the same inputs and time constraints.

Condition 3: Human with AI (HMT)

The LLM generates draft products, and the analyst reviews, edits, and approves them before finalization. The system captures the full edit trail: which sections were modified, what was added, what was removed, how confidence language was adjusted, and how long the review took. This condition tests the central hypothesis of the architecture: that human AI collaboration produces better products than either working alone.

Six evaluation metrics are applied across all three conditions. The first three are adopted from the evaluation triad proposed by Tellache et al [24]. *Answer relevance* (does the product address what the organization's intelligence requirements actually ask for), *context relevance* (is the retrieved and cited evidence appropriate to the finding), and *groundedness* (are claims traceable to specific source material rather than generated from parametric knowledge alone).

The remaining three are specific to this architecture. *Edit distance* measures how much analyst revision was needed to bring an AI generated draft to acceptable quality, applicable only to Condition 3. *Product usefulness* captures consumer feedback on whether the product was actionable, timely, and relevant to their decision making needs. *Tradecraft compliance* assesses whether the product meets minimum standards for source attribution, confidence expression using estimative language, and evidence traceability, drawing on the criteria identified by Bjurling and Raza [7] and the quality dimensions Kotsias et al. adapted from FM 2 0 [6].

Metric	AI	Human	HMT
Answer relevance (1-5)	3.2	4.1	4.5
Context relevance (%)	71	89	85
Groundedness (%)	62	94	88
Edit distance (%)	N/A	N/A	34
Product usefulness (1-5)	2.4	3.8	4.3
Tradecraft compliance (0-100)	38	82	91
Production time (min)	4	95	35

Fig. 8. Sample 7 Metric Evaluation Framework

The evaluation uses publicly available CTI data rather than synthetic test cases. Published threat reports from government CERTs, vendor advisories from organizations like Mandiant and CrowdStrike, and MISP community event data provide the input corpus.

C. Failure Analysis

The evaluation is designed to surface failures alongside successes. If the architecture does not produce better intelligence products than manual workflows, that result is itself a contribution because it identifies where current AI capability reaches its limits in CTI workflow integration.

Failure analysis focuses on three categories. The first is **generation failures**, cases where the LLM produces claims that are not grounded in retrieved evidence, misattributes threat activity, overstates confidence, or generates plausible sounding content that an analyst must fully rewrite. These failures map to the failure taxonomy documented by Meng et al., specifically spurious correlations, contradictory knowledge, and constrained generalization [20]. The evaluation records which category each failure falls into and at which pipeline stage it originated.

Context failures are cases where the organizational context layer fails to surface relevant findings or surfaces irrelevant ones. If the system consistently ranks low priority threats above high priority ones for the configured organization, or if it misses findings that an analyst would immediately flag as relevant, the context retrieval mechanism needs adjustment. These failures are measurable through the gap between what the system recommended for inclusion and what the analyst actually selected during the review step.

The third category is **product failures**, cases where the generated intelligence product meets technical accuracy requirements but fails to be useful. A strategic brief that restates threat data without translating it into executive decision points. An operational brief that identifies ATT&CK techniques correctly but does not connect them to the organization's specific asset exposure. A product that uses appropriate estimative language but buries the actionable finding under contextual background. These are the hardest failures to measure because they require evaluator judgment rather than automated scoring. The consumer feedback mechanism and tradecraft compliance assessment are designed to catch them.

Mezzi et al.'s finding that LLM performance degrades on realistic, full length CTI reports [28] suggests that generation failures will be more frequent when the prototype processes real world inputs rather than curated excerpts. The evaluation is designed to use real inputs precisely because sanitized benchmarks would understate the failure rate and overstate the architecture's readiness for operational deployment.

If the prototype demonstrates that HMT produced products score higher on usefulness and tradecraft compliance than either AI only or human only products, the architecture's core hypothesis is supported. If it does not, the failure data identifies which components need redesign and which assumptions were wrong. Either outcome advances the field's understanding of where AI fits within the CTI lifecycle.

VII. DISCUSSION & CAVEATS

The architecture proposed in this paper is a reference model. It addresses a gap in the CTI lifecycle, but it operates within constraints that are worth stating directly.

A. Constraints on AI in CTI Workflows

LLM reliability is the most immediate constraint. Mezzi et al. [28] and Meng et al. [20] have documented that models degrade on realistic inputs, produce overconfident assessments, and exhibit failure modes specific to CTI: spurious correlations, contradictory knowledge, and constrained generalization. The analyst review mechanism in Section V.E is built to catch these failures after they occur. It does not prevent them at generation time. The underlying reliability problem remains a design constraint that any implementation must account for.

Adversarial manipulation introduces a separate category of risk. Siskind examined how propaganda can infiltrate LLM outputs through training data and citation mechanisms, producing responses that appear balanced while systematically favoring particular narratives [18]. In a CTI context, this means generated intelligence products could surface adversarial framing of geopolitical threat activity without the analyst recognizing the bias. H. Huang et al. studied LLM generated misinformation in CTI specifically and found that fabricated threat intelligence can mislead attribution, delay incident response, and poison training data used by automated defenses [25]. Their finding that human analysts cannot reliably detect fabricated CTI without specialized training is a direct challenge to architectures that rely on analyst review as the primary quality control mechanism.

Shafee et al. demonstrated that adversarially crafted threat intelligence text can pass automated classification with a 97% false positive rate, flood analyst dashboards with fabricated intelligence, and poison retraining cycles [26]. Recent reporting documents these risks materializing in practice. The Bissa Scanner campaign used AI assisted techniques for mass exploitation and credential harvesting at scale, demonstrating that adversarial use of AI in offensive operations is an active and evolving threat [54]. A CTI pipeline that ingests data from open sources is exposed to this attack surface. The architecture's organizational context layer provides some filtering through relevance based triage, but it is not designed as an adversarial robustness mechanism.

Position bias affects how information is weighted during product generation. LLMs systematically underweight information appearing in the middle of long input contexts,

favoring content near the beginning and end [41]. For intelligence products generated from long CTI feeds containing multiple reports and advisories, some threat data may be deprioritized based on its position rather than its actual relevance. The organizational context layer partially addresses this by filtering inputs before they reach the generation stage, reducing length and concentrating the context. Position bias within the filtered input remains a factor.

B. Open Questions

How to evaluate intelligence product quality beyond NLP metrics remains the most fundamental open question. The evaluation framework in Section VI introduces product usefulness and tradecraft compliance, but both rely on human judgment. A scalable, repeatable quality metric for CTI intelligence products does not yet exist. Bjurling and Raza identified the gap [7]. Closing it remains work for the broader field.

Whether the architecture transfers across organizational contexts is an empirical question this paper cannot answer. The evaluation uses a hypothetical financial services organization. A healthcare system, a government agency, a small MSSP, and a critical infrastructure operator each present different requirements, regulatory environments, and analyst populations. Whether the organizational context layer adapts or requires redesign for each is unknown without testing.

The active monitoring boundary is worth stating explicitly. The architecture sits after detection and collection, operating on data that has already been gathered and structured. It does not attempt to address real time network monitoring or active threat detection, where traditional filtering and signature based methods remain more effective.

C. Research Limitations

The literature review is bounded by the databases searched, the keywords used, the inclusion criteria applied, and the English language. Papers published after the search, proprietary vendor research, and classified government work are not represented. The lifecycle coverage matrix reflects the author's classification judgments. A different reviewer might assign different coverage ratings to the same papers.

The architecture is proposed and justified but not fully implemented. The prototype in Section VI tests a subset of the components. Compliance context mapping in its full form awaits future implementation. Claims about the architecture's effectiveness are conditional on empirical results this paper does not yet provide.

VIII. FUTURE WORK

The following directions build directly on the architecture and evaluation framework described in this paper.

Empirical Validation.

The most immediate next step is implementing and testing the full architecture. The prototype described in Section VI evaluates a subset of the components against a constrained organizational profile. A complete implementation covering all three intelligence product classes, the compliance context mapping layer, and the anonymization pipeline would allow evaluation against the full architecture. It would also surface engineering constraints that a reference model intentionally abstracts away: latency, token limits, retrieval scaling, and context window management.

Cross Organizational Testing

The evaluation in this paper uses a hypothetical financial services organization. Whether the architecture generalizes is unknown without testing across different sectors and regulatory environments. A healthcare system under HIPAA, a government agency under FISMA, a small MSSP serving multiple clients, and a critical infrastructure operator each present different intelligence requirement structures and analyst populations. Whether the organizational context layer adapts or requires redesign for each is an open question.

Adversarial Robustness

The discussion in Section VII documents known attack vectors against AI CTI systems, including adversarially crafted threat text [26], propaganda leakage [18], and fabricated intelligence [25]. How the architecture's context filtering, analyst review, and source provenance surfacing perform against these attacks under controlled conditions is a research question worth isolating.

Reporting Quality Metrics

Current benchmarks measure extraction accuracy. Developing scalable, repeatable quality metrics for CTI intelligence products would benefit the field independently of this architecture. Bjurling and Raza identified the gap between extraction benchmarks and tradecraft standards [7]. A validated evaluation instrument for reporting quality remains an open contribution.

Trust Calibration

The architecture's feedback instrumentation (edit tracking, consumer response data, revision density) makes longitudinal study of analyst AI trust possible. How trust develops over time, whether it follows the developmental trajectory documented in human AI team research [39], and how it

responds to system failures are questions that require sustained deployment rather than a single evaluation.

IX. CONCLUSION

This paper examined how AI capabilities distribute across the CTI lifecycle by mapping 54 papers against the six phases of the SEI Cyber Intelligence Framework. The analysis found a consistent pattern: research and tooling concentrate in data gathering and threat analysis while strategic analysis, reporting and feedback, and human machine teaming remain underserved. Benchmark evaluations corroborate this, showing that model performance degrades as tasks move from structured extraction toward reasoning, synthesis, and product generation. Practitioner evidence confirms that the operational bottleneck in CTI programs sits at contextualization, prioritization, and communication rather than at collection.

The reference architecture proposed here addresses these gaps by treating intelligence product generation as a primary design target. It contextualizes threat data against organizational intelligence requirements, generates operational, strategic, and external product classes from a shared evidence base, involves the analyst as a co-equal partner through structured edit sessions, and captures feedback that flows back into collection priorities. Compliance context mapping and anonymization for external sharing extend the architecture into regulated industries and information sharing communities where current tools offer limited support.

The architecture is a reference model. Empirical validation through the three condition evaluation described in this paper is the necessary next step, testing whether the reporting-centered workflow produces more useful and tradecraft compliant intelligence products than current approaches. Those results, whether they confirm or challenge the design, will determine which components survive contact with real analyst workflows and contribute reporting focused evaluation metrics to a field that has measured extraction accuracy far more than product usefulness.

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